Position–Time Graphs: Straight-Line Motion (A–H)

Worked solutions · Supplementary answer key prepared for Lesson 3. Use only after completing the worksheet.

For each graph, the straight-line gradient gives constant velocity. Positive and negative positions are signed relative to the chosen origin.

Graph A

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (20 m – (0 m)) / (5 s – 0 s)

Answer: v = +4 m/s

Interpretation: Position increases uniformly; the velocity is constant in the positive direction.

Graph B

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (10 m – (0 m)) / (5 s – 0 s)

Answer: v = +2 m/s

Interpretation: Position increases uniformly, at a lower velocity than in A.

Graph C

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (20 m – (10 m)) / (5 s – 0 s)

Answer: v = +2 m/s

Interpretation: Starts at a positive position and moves at constant positive velocity.

Graph D

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (10 m – (10 m)) / (5 s – 0 s)

Answer: v = 0 m/s

Interpretation: Position remains constant; the object is stationary.

Position–Time Graphs: Straight-Line Motion (continued)

Graph E

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (-15 m – (0 m)) / (5 s – 0 s)

Answer: v = –3 m/s

Interpretation: Position decreases; the object moves at constant negative velocity.

Graph F

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (0 m – (15 m)) / (5 s – 0 s)

Answer: v = –3 m/s

Interpretation: Moves at constant negative velocity towards the origin.

Graph G

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (-12 m – (-12 m)) / (5 s – 0 s)

Answer: v = 0 m/s

Interpretation: Position remains negative and constant; the object is stationary.

Graph H

Equation: $v = \Delta x / \Delta t = (x_f - x_i) / (t_f - t_i)$

Rearrangement: Not required; velocity is already the subject.

Substitution: v = (-3 m – (-15 m)) / (5 s – 0 s)

Answer: v = +2.4 m/s

Interpretation: Position increases from a negative value; the velocity is positive.